Panskura Banamali College (Autonomous)

B.Sc.(H) 5th Semester Examination 2022 (Online mode)

Subject : Computer Science **Paper: DSE2T**

Paper Name : Numerical Computing

F.M. : 40 **Time : 2 Hours**

Answer any four questions: **4×10 = 40**

1. i) Prove that $\Delta^n f(x) \cong h^n f^n(x)$, where the notations have their usual meanings. Use finite difference method to find the values of a and b in the following table.

x	0	2	4	6	8	10
f(x)	-6	a	10	b	30	40

ii) Find the absolute, relative and percentage error in x_A when $x_T = \frac{2}{3}$ and $x_A = 0.6666$. Write the differences between relative error and absolute error. (2+5)+3

2. i) Show that an iteration scheme to compute $\sqrt[k]{a}$ is given by $x_{n+1} = \frac{1}{k}[(k-1)x_n + \frac{a}{{x_n}^{k-1}}]$.
ii) Can we derive Trapezoidal rule of numerical integration from general quadrature formula based on Newton's forward difference interpolation formula? If so, derive this. Evaluate $\int_0^3 (x^2 - 2x)dx$ by this rule taking 12 intervals. Write the degree of precision of this rule. 3+(1+1+4+1)

3. i) Find a root of the equation $x^2 + 7\cos x - 2 = 0$ by bisection method, correct up to three decimal places.
ii) Show that $\sum_1^n \frac{w(x)}{(x-x_i)w'(x_i)} = 1$, where $w(x) = (x - x_0)(x - x_1) \dots (x - x_n)$. Prove that the terms used in this summation are invariant. 4+(4+2)

4. i) Solve the equations $27x + 6y - z = 27, 6x + 15y + 2z = 36, x + y + 54z = 55$ correct up to four decimal places by Gauss-Jacobi's method. Write the sufficient condition of convergence of this method.
ii) State Simpson's 1/3 rule for numerical integration. Use its composite form to evaluate $\int_0^3 (2x - x^2)\,dx$ taking 12 intervals. (3+3)+(1+3)

5. i) Use Newton-Raphson method to find a root of the equation $xe^x - cos\ x = 0$ correct up to three decimal places. Prove that this method converges quickly than the method of iteration.
ii) Solve the system of equations $2x + y + z = 8,\ x - y + 2z = 4, 2x + 2y - z = 6$ by using Gauss-elimination method. (5+1)+4

6. i) State the Newton – Gregory backward difference interpolation formula. Using it and the following table find the value of **y** at **x = 0.612**.

X	0.61	0.62	0.63	0.64	0.65
Y	1.840431	1.858928	1.877610	1.896481	1.915541

ii) Is Euler's method a single-step explicit method? Using this method find the values of y(0.1) and y(0.2) from the differential equation $\frac{dy}{dx} = x^2 + y^2$ with y(0)=1. (1+4)+(1+4)

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